

Lec 13 : 18th Feb 2026 :

Regularization :

Manifold hypothesis : data that we get, is a very very small manifold in the ambient space.

→ Classifier is a bifurcator in a small manifold in the ambient space.

→ any method that increases bias is Regularization.

→ choice of class condⁿ is dealing with bias.

→ Any architectural choice is a bias.

→ Ways of Regularization

1) Parameter penalty

2) Architectural restrictions.

3) Optimization procedures. (Early stopping)

⇒ Regularization via Parameter Penalty:

Recall ERM

$$\hat{h}_0^*(x) = \underset{h_0 \in \mathcal{H}}{\operatorname{argmin}} \quad \frac{1}{n} \sum_{i=1}^n \angle(h_0(x_i), y_i)$$

Regularized ERM.

Define a func $\theta: \rightarrow \mathbb{R}^+$

Reg-ERM.

$$\begin{aligned} h_{\theta}^*(x) = \underset{h_{\theta}}{\operatorname{argmin}} \quad & \frac{1}{n} \sum_{i=1}^n L(h_{\theta}(x_i), y_i) \\ \text{s.t} \quad & \Omega(\theta) \leq k \\ \text{where} \quad & k \in \mathbb{R}^+ \end{aligned}$$

To Solve Reg-ERM

$$\hat{R}_{\lambda}(\theta) = \frac{1}{n} \sum_{i=1}^n L(h_{\theta}(x_i), y_i) + \lambda (\Omega(\theta) - k)$$

$$h_{\theta}^* = \underset{\theta}{\operatorname{argmin}} \hat{R}_{\lambda}(\theta)$$

λ : Reg Cont : hyper parameter.

* use the Gradient Based numerical methods to solve the Reg-ERM. (some times

choices for $\Omega(\theta)$:

i) $\Omega(\theta) = \|\theta\|_p$

$\|\theta\|_2$: L2-Reg : Ridge

$\|\theta\|_1$: L1-Reg : Lasso

ii) $\Omega(\theta) = \exp(-\theta)$

Why should this increase Bias:

Show that the Bias of Reg model is always more than Un Reg model : Do it in Tutorials

Equivalence b/w Reg-ERM and MAP Estimates:

$p(\theta|x) \propto p(x|\theta) \cdot p(\theta)$: map estimate

Consider an example, where

$$p_{y|x} = p_v \sim N(y_i; h_\theta(x_i), I)$$

$$p_{(\theta|v)} \propto p_{(v|\theta)} p_\theta$$

$$h_\theta \approx -\frac{1}{n} \sum_{i=1}^n \|y_i - h_\theta(x_i)\|^2$$

Suppose $p_{\theta} = N(\theta; 0, I)$

$$p_{\theta|v} \propto p_{v|\theta} p_{\theta}$$

$$\ln p_{\theta|v} \propto \ln p_{v|\theta} + \ln p_{\theta}$$

$$\propto -\frac{1}{n} \sum_{i=1}^n \|y_i - h_{\theta}(x_i)\|^2 - \|\theta\|_2^2$$

In Prob treatment

$$\theta^* = \operatorname{argmax}_{\theta} l(\theta)$$

$$= \operatorname{argmin}_{\theta} -l_{\theta}(\theta)$$

$$= \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n \|y_i - h_{\theta}(x_i)\|^2 + \|\theta\|_2^2$$

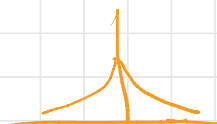
This is same as Reg ERM, with $\Omega(\theta) = \|\theta\|_2^2$

This shows that

Reg ERM with $\|\theta\|_2^2$ is same as Bayesian estimate with a Gaussian Prior.



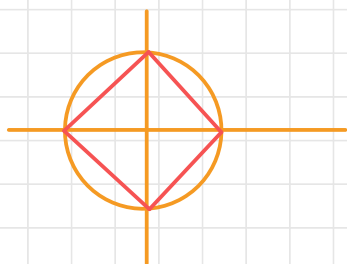
Gaussian prior



Laplacian prior

induces
sparsity

Suppose $p_0 \sim \text{laplacian}(\alpha)$, then prob of parameters around mean will be higher than that of Gaussian.
 show in Tutorial.



- L_2 norm
- L_1 norm

Another example for Regularization:

Suppose $D = \{(x_i, y_i)\}_{i=1}^n \sim \text{iid } P_{xy}$

Create a new Dataset as follow.

$$\forall x_i \in D$$

$$\hat{x}_i = x_i + \epsilon \quad \epsilon \sim N(0, \sigma^2)$$

$$\hat{D} = \{(\hat{x}_i, y_i)\}_{i=1}^n$$

ERM on $\hat{D}$ is equivalent to $\equiv$ Reg ERM on D

Proof in Tutorial
 how is α & σ related?

Revisiting Gradient Descent:

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(h_{\theta}(x_i), y_i)$$

Gradient descent

$$\theta^{t+1} \leftarrow \theta^t - \beta \nabla_{\theta} \hat{R}(\theta)$$

Let's alter the Gradient descent.

$$\text{Define } \hat{R}_B(\theta) = \frac{1}{B} \sum_{i=1}^B \ell(h_{\theta}(x_i), y_i)$$

where $B \ll N \rightarrow x_i \in D_B$

where $D_B \subset B$ -length subset of D .

$$\theta^{t+1} \leftarrow \theta^t - \beta \nabla_{\theta} \hat{R}_B(\theta)$$

The above is Stochastic Gradient descent (SGD)

$\hat{R}_B(\theta)$ is a noisy estimate on $\hat{R}(\theta)$

$\rightarrow$ perturb of Gradient has a Regularization effect

$\rightarrow$ Perturb data also has a Regularization effect

Similar to perturbation of Gradient.

Lecture 14: 23rd Feb 2026;

Max-Margin Classifier : (a) Support Vector Machine.

Consider a Binary Classification problem

$$X \in \mathbb{R}^d \text{ \& \; } y \in \{-1, 1\}$$

Discriminant func:

Recall that any Binary Classification problem, we have.

$$h(x) = \begin{cases} 1 & g(x) > 0 \\ -1 & \text{o/w} \end{cases}$$

where $g(x)$: func of x

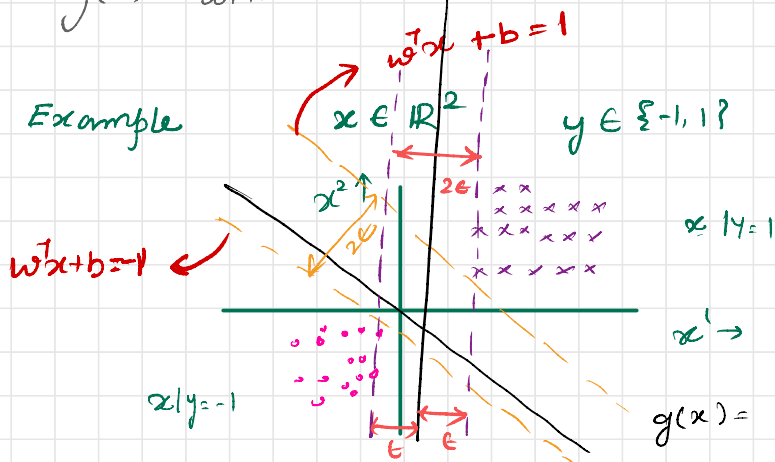
$$g(x) : \mathbb{R}^d \rightarrow \mathbb{R}$$

Discriminant func

* For some specific cases of class condⁿ density

$g(x)$ will be linear

Example



Linear Separability:

A Dataset D is linearly separable s.t

$$w^T x_i + b > 0 \quad \forall \quad x_i : y_i = 1$$

$$w^T x_i + b < 0 \quad \forall \quad x_i : y_i = -1$$

→ if data is linearly separable, $\exists$ ∞ discriminator func.

question: is there an optimal discriminat func
among all these?

* Lets consider optimality via maximizing the margin

* Consider linearly separable data.

Since its lin separable $\exists \epsilon > 0$ s.t

$$w^T x_i + b > \epsilon \quad \forall \quad x_i : y_i = 1$$

$$w^T x_i + b < -\epsilon \quad \forall \quad x_i : y_i = -1$$

* Now, w & b can be rescaled s.t

$$\hat{w}^T x_i + \hat{b} \geq 1$$

$$\hat{w}^T x_i + \hat{b} \leq -1$$

notationally easy to write
 w & b .

$$w^T x_i + \hat{b} \geq +1 \quad \forall x_i : y_i = 1$$

$$w^T x_i + \hat{b} \leq -1 \quad \forall x_i : y_i = -1$$

* we can combine both these and can write

$$y_i (w^T x_i + b) \geq 1 \quad \forall i$$

* For linear separable case, $\exists w$ & b s.t

$$y_i (w^T x_i + b) \geq 1 \quad \forall i = 1, \dots, m.$$

also $\exists$ no data points b/w the hyperplanes.

$$w^T x + b = 1 \quad \& \quad w^T x + b = -1$$

and these are parallel to $w^T x + b = 0$

* Consider the perpendicular distances b/w the

$$\text{margin.} : \frac{2}{\|w\|}$$

* One criterion for optimality of $g(x)$, is that

$$\frac{2}{\|w\|_2} \text{ has to be maximized.}$$

Mathematically

$$\min_w \frac{\|w\|}{2}$$

$$\text{s.t. } (w^T x_i + b) y_i \geq 1 \quad \forall i$$

ERM, linear hypothesis, hinge loss & L_2 Reg is SVM

The above is a constrained optimization problem with convex cost & linear constraints.

* Solve the above via KKT optimizations.

Define Lagrangian for the above problem.

$$L(w, \mu, b) = \frac{\|w\|}{2} + \sum_{i=1}^n [1 - y_i (w^T x_i + b)] \mu_i$$

KKT for the above problem:

$$a) \nabla_w L(w^*, \mu^*) = 0$$

$$b) \mu_j^* \geq 0$$

$$c) \mu_j^* (1 - y_j (w^{*T} x_j + b)) = 0$$

a) $\nabla_w L = 0$ algebra

$$w^* = \sum_{i=1}^n \mu_i^* y_i x_i \rightarrow \text{lin comb of data points}$$

$$\nabla_b L = 0 \quad \sum_{i=1}^n \mu_i^* y_i = 0$$

b) $\mu_i^* \geq 0 \quad \forall i$

Define $S = \{ x_i : \mu_i^* > 0 \}$

$$\Rightarrow w^* = \sum_S \mu_i y_i x_i$$

* To get w^* , we need optimal Lag multipliers μ_i^*

* To find μ_i^* , we solve the dual problem.

* Duality of optimization:

$$L(w, \mu, b) = \frac{w^T w}{2} + \sum_{i=1}^n [1 - y_i (w^T x_i + b_i)] \mu_i$$

Define a Dual func

$$q: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$q(\mu) = \inf_w L(w, \mu, b)$$

$\inf$: Greatest Lower bound.

min v/s $\inf$
min can be found $\swarrow$
 $\searrow$ min exists & can be found.

Construct Dual optimization problem:

$\rightarrow$ primal-min then dual is max & Vice versa

$$\rightarrow \max_{\mu} q(\mu)$$

$$\text{s.t } \mu_i \geq 0 \quad \forall i$$

To solve this we need primal dual relationship:

w^* & μ^* are optimizers for primal & dual respectively iff

$\Rightarrow w^*$ & μ^* are feasible for primal &

dual respectively.

$$\begin{aligned} 2) \quad L(\omega^*, \mu^*) &= \min_{\omega} q(\mu^*) \\ &= \min_{\omega} L(\omega, \mu^*) \end{aligned}$$